UNIVERSITY OF CAMBRIDGE

Application for Employment

In Progress

Applicant Name	Zhanhong He
Position applied for	Research Associate
Department	Faculty of Music
Vacancy reference	GT51091: Research Associate

PERSONAL DETAILS

Last name	He
First name(s)	Zhanhong
Title	Mr
Current address	U5/5 Clark Street Crawley Western Australia 6009 Australia
Primary phone number	+852 54287456
Secondary phone number	+61 410128422
Email address	zhanh.he.uwa@gmail.com
Immigration status	Are you a settled worker? **No** If you are not a settled worker, do you already have permission to work in the UK? **No** If you already have permission to work in the UK, please specify your permission type and
UK NI Number	
Availability or Notice Period	None

REFERENCES

1) Individual Referee

Name	Professor Roberto Togneri
Position	Retired Professor, PhD supervisor

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Address	University of Western Australia 35 Stirling Highway Crawley Western Australia 6009 Australia
Telephone number	
E-mail address	roberto.togneri@uwa.edu.au
At what point in the recruitment process may references be gathered?	At any point in the process

2) Individual Referee

Name	Professor Yaolong Ju
Position	Assistant Professor
Address	Great Bay University No. 16 Daxue Road, Songshan Lake High-Tech Industrial Development Zone Dongguan Guangdong 523808 China
Telephone number	
E-mail address	juyaolong@gbu.edu.cn
At what point in the recruitment process may references be gathered?	At any point in the process

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APPLICANT DECLARATION & DATA CONSENT

When you are ready to submit your final application, you must agree to the declaration below and click Submit Application.

By doing so, you are confirming that:

- You understand how the University will use and store your personal data, having read the section on Applicant Data on our HR web pages.
- The information you have given in your web-based application for employment and any supporting documents is correct and complete.
- I understand that the documents I have uploaded to support my application have been converted to pdf format, and I confirm that I have checked them and they are an accurate representation of the originals.
- You understand that you will not be able to make any changes after submitting your application.
- You understand that failure to disclose any relevant information or the provision of false information may lead to dismissal/withdrawal of any offer of employment made to you.
- You understand that the University of Cambridge may check all or any of the information provided as part of your application or given by your referees.
- You understand that an appointment, if offered, will be subject to the receipt of references, and the outcome of any relevant pre-employment checks, which the University regards as satisfactory.

Signature: Date:

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ZHANHONG HE



PhD, Dec 2026 Graduation, University of Western Australia (UWA Profile)

zhanh.he.uwa@gmail.com

https://github.com/zhanh-he/ | https://www.linkedin.com/in/zhanhong-he/

Research Profile

My PhD research focuses on expressive musical dynamics, audio-symbolic modelling and music generation, including MIDI velocity estimation, score-informed transcription refinement, multitask learning and human perceptual evaluation. My industrial research experience at Dolby and Huawei spans music generation, reinforcement learning and spatial audio. This is complemented by university teaching and project co-supervision in electronic engineering and computer science. My music coursework and practical music-making experience also motivate my interest in developing tools that give musicians intuitive control over musical expression.

Education

University of Western Australia

PhD research in Electrical and Electronic Engineering | February 2022-December 2026

Doctoral defense completed; graduation certificate expected November 2026. Formal award date pending confirmation.

Research focus: musical dynamics, audio and symbolic music processing, and machine learning.

University International Fee Scholarship; University Postgraduate Awards; 2026 UWA Data Institute Award.

University of Western Australia

BSc (Honours) in Engineering Science | October 2018-December 2021

First Class Honours; GPA 7/7; WAM 87.7/100.

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Selected Research

Expressive dynamics and audio-symbolic modelling (University of Western Australia)

- Developed score-informed Transformer models for MIDI velocity refinement and multitask CRNNs for jointly estimating musical dynamics, beats, downbeats and structural changes.
- Adapted piano-pretrained velocity estimation to other instruments using differentiable SoundFont proxies and loudness-based supervision, reducing reliance on note-level velocity labels.

LLM post-training for music generation (Huawei Hong Kong)

- Applied DPO and GRPO to LLM-based music generation, including vocal-to-accompaniment and lyrics-to-music tasks
- Developed music-specific reward signals and rollout workflows, and collected and analysed human-preference data for model evaluation and post-training.

Choral transcription and agentic data workflows (Great Bay University)

- Conducted research on choral singing transcription, including note transcription and vocal-part assignment.
- Built agentic workflows for data collection and dataset construction, incorporating ACE Studio synthesis.

Generative modelling and perceptual evaluation (Dolby Australia)

- Explored U-Net, VAE and masked Transformer architectures for music generation, including a U-Net approach to per-note MIDI velocity prediction. Designed a listening test with professional listeners.

Research Experience

Huawei Hongkong Research Centre - Research Intern, full time | Jun-Dec 2026

- Conducted research on music-model post-training and human-preference evaluation.

Dolby Laboratories, Sydney - Research Intern, full time | May-September 2024

- Developed music-generation models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted professional listening tests to evaluate generated musical expression.

Great Bay University - Visiting Student Researcher, part time | Feb-Dec 2026

- Explored pretrained music models (e.g.,MERT) and agentic workflows for music research.

Teaching And Collaboration

University of Western Australia - Tutor and Lab Facilitator, part time | Feb 2023-Dec 2025

- Supported ELEC4404 Signal Processing and GENG2000 Engineering Practices.
- Co-supervised CITS4011 Computer Science Honours projects and final-year designs.

Foreign Trade University, HCMC, Vietnam – Teaching Assistant, part time | Jun 2019-Jul 2019

- Held 2-month Language and Culture Summer Program for local students.

Publications And Preprints

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 Zhanhong He, Hanyu Meng and Yaolong Ju. CASM: Context-Aware Semi-Markov Post-Processor for Beat Tracking. Under review in ICASSP2027. [Demo] [Arxiv]

 Hanyu Meng, **Zhanhong He**, Zixun Guo and Yaolong Ju. Toward Part-Aware Choral Transcription with singing voice assignment. Under review in ICASSP2027. [Demo] [Arixv]

 Hanyu Meng, Hao Shi, **Zhanhong He**, et al. Speaker-Consistent Grounding for Confusion-Resilient LLM-Based Target Speech Extraction. Under review in ICASSP2027. Mingda Song, Yueqiao Zhang, Yulun

 Yulun Wu, **Zhanhong He**, et al. HCMAMBA: Harmonically Informed State-Space Modeling for Large-Vocabulary Audio Chord Recognition. Under review in ICASSP2027.

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 Zhanhong He, Hanyu Meng, Jiahao Zhao, Ganghui Ru and Yaolong Ju. MIREX 2026 Submission: Context-Aware Semi-Markov Post-Processor for Beat and Downbeat Estimation. ISMIR Late breaking Demo, 2026. [MIREX]

 Hanyu Meng, **Zhanhong He** and Yaolong Ju. PawChorale: A Scalable Synthetic Dataset for Part-Aware Choral Transcription and Separation. ISMIR Late breaking Demo, 2026. [Dataset]

 Zhanhong He, Hanyu Meng, David Defeng Huang and Roberto Togneri. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. Int. Soc. Music Info. Retrieval (ISMIR), 2026. [GitHub]

 Zhanhong He, Hanyu Meng, Defeng (David) Huang and Roberto Togneri. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP), 2026. [GitHub]

 Zhanhong He, David Cooper, Defeng Huang and Roberto Togneri. Filling MIDI Velocity using U-Net Image Colorizer. Int. Symposium on Computer Music Multidisciplinary Research (CMMR), 2025. [GitHub]

Music Practice and Interactive Systems

- **UWA Conservatorium of Music** coursework in electronic music, music generation and ensemble performance, 2023; composition and mixing music with Ableton Live [Demo Video], and participation in a public ensemble concert [YouTube].
- Built a coursework HMM system for automatic piano and environmental-sound generation system, using Max/MSP with Ableton Live.
- Built an Arduino & PCB-based theremin with a MIDI interface, programmed in C, at a Physical Interaction Design for Music workshop, Stanford University CCRMA, 2023.

Technical Skills

Python and PyTorch; LLM Post-Training; DDSP, U-Net and VAE research; audio and symbolic music processing; human perceptual evaluation; Ableton Live and Max/MSP.

Languages: English (fluent); Chinese and Cantonese (native).